

Broad Technology Space – Flow Cytometry Core

Sample Preparation & Cell Sorting Guidelines v.1.0

Sony SH800 & MA900

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Overview

These guidelines apply to cell sorting on the Sony SH800 and MA900 sorters in the BTS Flow Cytometry Core. Please read this document carefully before your sort appointment. If you have questions about your specific experiment, reach out to core staff ahead of time; we are happy to help you plan.

Training Requirements

To gain access to the sorters, all users must attend a Sony sorter training session and complete a supervised sort. After your training session, email core staff to schedule your supervised sort. The goal of the supervised sort is to confirm proficiency in operating the instrument independently; not to execute a full experiment. Keep it simple; a small pilot sort or a compensation test are good examples of appropriate supervised sort experiments.

Scheduling

Once trained you'll be able to book time on the sorters through Bookit (<https://core.bookitlab.com/broad/Login>). New users can schedule a consultation before you sort so we can advise on instrument selection, nozzle size, and sample requirements. After completing your training and supervised sort, you will have 24/7 access to the Sony sorters. Core staff are happy to assist anyone with their sort, we just ask for a minimum of 24 hours notice ahead of your booking to guarantee availability.

We understand that sample prep is demanding and things don't always go as planned. However, please be considerate of other users and stick to your reserved time. While we can occasionally help accommodate scheduling conflicts, we ask that this not become a regular occurrence; other researchers depend on their reserved time as much as you do. If you need more time than booked, please adjust your reservation in Bookit.

Start-up and Shutdown

The first user of the day is responsible for instrument startup and calibration; the last user of the day is responsible for shutdown. If you need staff to cover startup or shutdown, we are happy to help when available, please let us know at least 24 hours in advance.

Nozzle Size

Sorting chips come in nozzle sizes of 70µm, 100µm, and 130µm. Nozzle size must be 5 to 7x larger than particle size. The 100µm chip is the standard, "one-size-fits-all" chip size. Talk to core staff if you are unsure about which chip to use.

Sample Preparation

Single-Cell Suspension

All samples must be in a true single-cell suspension. Clumps are the most common cause of sort failure or nozzle/sample-line clogging.

- Use a buffer to reduce cell aggregation. Example: Ca²⁺/Mg²⁺-free buffer with 0.5% BSA or 2% FBS, 2 mM EDTA, and 25 mM HEPES (pH 7.0).

- If dissociating tissue, confirm single-cell suspension visually under a microscope before stopping the reaction.
- For samples with low viability, consider adding **25–50 µg/mL DNase I with 5 mM MgCl₂** (without EDTA) to prevent clumping from extracellular DNA.

Cell Concentration

We recommend a **maximum** final concentration of 5×10^6 cells/mL for most sorting applications. Bring an appropriate buffer to dilute if your sample is more concentrated. For single-cell sorting (e.g., into plates), use the lower end of this range.



Filtering

All samples must be filtered immediately before sorting to prevent clogging. Each instrument station is stocked with Celltrix nylon-mesh filters with 30µm or 50µm pore sizes. For sorting with a 100µm chip, a 30µm filter is recommended.

Viability

Dead cells and debris negatively impact sort purity and clog the instrument. Aim for >80% viability before sorting. Include a live/dead discriminator (e.g., DAPI, 7-AAD, or viability dye) in your panel whenever possible.

Controls

Proper controls are required for every sort. Please bring all of the following:

- **Unstained/Parental:** Unlabeled cells with no antibody, dye, or fluorescent protein. Bring at least 500 µL at >100,000 cells if this is a new experiment.
- **Single-color controls:** One control for each fluorochrome, dye, or fluorescent protein in your panel. Compensation beads are acceptable for antibody channels.

Optional:

- **FMOs:** Stain with all fluorochromes except one. Critical for setting gates on dim or continuous markers.

Note: Do NOT add live/dead dye to your unstained or single-color controls. We recommend titrating all antibodies and viability dyes.

Tubes & Collection

Sample Tubes

The 5 mL FACS tubes provided are polystyrene (clear); for rare populations, polypropylene tubes (opaque/frosted) are recommended to minimize cell loss. Do not use filter-cap tubes; these leave the sample exposed to contamination.

- Acceptable: 1.5 mL microcentrifuge, 5 mL round-bottom FACS tube, or 15 mL conical tubes
- Keep samples on ice and protected from light where appropriate

Collection Tubes

The SH800 and MA900 can collect into the following:

- 1.5 mL microcentrifuge tubes
- 5 mL polypropylene or polystyrene FACS tubes
- 15 mL conical tubes
- Plates: 384-, 96-, 48-, 24-, 12-, and 6-well culture plates; 96-well and 384-well PCR plates; 8-well strips

Collection Tube Coating

Consider coating collection tubes to prevent cells from sticking to tube walls, especially for low-yield sorts:

- ☐ Fill tubes with PBS + 20% FBS or 5% BSA
- ☐ Incubate 2 hours at room temperature or overnight at 4°C
- ☐ Just before sorting, empty and replace with your collection media

Note: Tubes should not dry out before the sort begins.

Sort Time

Sort time depends on several factors: your purity goal (efficiency vs. purity), your cell type (fragile or large cells require lower pressures and larger nozzles), and your sample concentration (more dilute samples take longer to process). Use the [Sony Sort Rate Calculator](#) to help you approximate sort times.

Collection Media

- Use growth media with high serum (FBS or BSA). Add antibiotics/antimycotics if culturing afterward. Fill 10%-20% of the tube volume — cells will be diluted by sheath buffer during the sort.

Sort Checklist

Bring all of the following to your sort appointment:

- ☐ Filtered sample in polypropylene tubes
- ☐ Unstained/Parental control
- ☐ Single-color compensation controls (one per fluorochrome)
- ☐ FMO controls as needed
- ☐ Pre-coated collection tubes or plates (bring extras)
- ☐ Dilution buffer
- ☐ Collection media
- ☐ Ice (if keeping samples cold)

Biosafety

Users must be aware of the biosafety level of the samples they are working with. The Biosafety Level assigned for cell sorting is often higher than that of bench work because sorting generates aerosols that can be inhaled. Users must follow standard BSL2 practices in the flow facility. Proper PPE is mandatory. We have a designated Sony MA900 (SuperWaffles) located in a separate room for BL2+ samples. Please notify core staff before your sort if your sample involves:

- Lentiviral vectors
- Infectious agents
- Live bacteria or yeast (These microorganisms pose a contamination risk and special preparations are required to sort on SuperWaffles; additional charges may apply. Please contact us to plan ahead.)